In [3]:
```
from pathlib import Path

project_dir = Path.home() / "Downloads" / "SCADA Water Analytics"
raw_dir = project_dir

print("Project folder exists:", project_dir.exists())
print("CSV files found:", len(list(raw_dir.glob("*.csv"))))

for file in sorted(raw_dir.glob("*.csv")):
    print(file.name)
```

```
Project folder exists: True
CSV files found: 10
2018_Leakages.csv
2018_SCADA_Demands.csv
2018_SCADA_Flows.csv
2018_SCADA_Levels.csv
2018_SCADA_Pressures.csv
2019_Leakages.csv
2019_SCADA_Demands.csv
2019_SCADA_Flows.csv
2019_SCADA_Levels.csv
2019_SCADA_Pressures.csv
```

In [21]:
```
import pandas as pd
demands_2018 = pd.read_csv(r'C:\Users\HP\Downloads\SCADA Water Analytics\2018_SCADA
```

```
---------------------------------------------------------------------------
ParserError                               Traceback (most recent call last)
Cell In[21], line 2
      1 import pandas as pd
----> 2 demands_2018 = pd.read_csv(r'C:\Users\HP\Downloads\SCADA Water Analytics\201
8_SCADA_Demands.csv')

File ~\miniconda3\envs\wateranalytics\Lib\site-packages\pandas\io\parsers\readers.p
y:872, in read_csv(filepath_or_buffer, sep, delimiter, header, names, index_col, use
cols, dtype, engine, converters, true_values, false_values, skipinitialspace, skipro
ws, skipfooter, nrows, na_values, keep_default_na, na_filter, skip_blank_lines, pars
e_dates, date_format, dayfirst, cache_dates, iterator, chunksize, compression, thous
ands, decimal, lineterminator, quotechar, quoting, doublequote, escapechar, comment,
encoding, encoding_errors, dialect, on_bad_lines, low_memory, memory_map, float_prec
ision, storage_options, dtype_backend)
    860 kwds_defaults = _refine_defaults_read(
    861     dialect,
    862     delimiter,
   (...)    868     dtype_backend=dtype_backend,
    869 )
    870 kwds.update(kwds_defaults)
--> 872 return _read(filepath_or_buffer, kwds)

File ~\miniconda3\envs\wateranalytics\Lib\site-packages\pandas\io\parsers\readers.p
y:306, in _read(filepath_or_buffer, kwds)
    303     return parser
    305 with parser:
--> 306     return parser.read(nrows)

File ~\miniconda3\envs\wateranalytics\Lib\site-packages\pandas\io\parsers\readers.p
y:1950, in TextFileReader.read(self, nrows)
   1943 nrows = validate_integer("nrows", nrows)
   1944 try:
   1945     # error: "ParserBase" has no attribute "read"
   1946     (
   1947         index,
   1948         columns,
   1949         col_dict,
-> 1950     ) = self._engine.read(  # type: ignore[attr-defined]
   1951         nrows
   1952     )
   1953 except Exception:
   1954     self.close()

File ~\miniconda3\envs\wateranalytics\Lib\site-packages\pandas\io\parsers\c_parser_w
rapper.py:238, in CParserWrapper.read(self, nrows)
    236 try:
    237     if self.low_memory:
--> 238         chunks = self._reader.read_low_memory(nrows)
    239         # destructive to chunks
    240         data = _concatenate_chunks(
    241             chunks, self._low_memory_column_labels(chunks)
    242         )

File pandas/_libs/parsers.pyx:832, in pandas._libs.parsers.TextReader.read_low_memor
y()
```

```
--> 832 'Could not get source, probably due dynamically evaluated source code.'

File pandas/_libs/parsers.pyx:897, in pandas._libs.parsers.TextReader._read_rows()
--> 897 'Could not get source, probably due dynamically evaluated source code.'

File pandas/_libs/parsers.pyx:868, in pandas._libs.parsers.TextReader._tokenize_rows
()
--> 868 'Could not get source, probably due dynamically evaluated source code.'

File pandas/_libs/parsers.pyx:885, in pandas._libs.parsers.TextReader._check_tokeniz
e_status()
--> 885 'Could not get source, probably due dynamically evaluated source code.'

File pandas/_libs/parsers.pyx:2090, in pandas._libs.parsers.raise_parser_error()
-> 2090 'Could not get source, probably due dynamically evaluated source code.'

ParserError: Error tokenizing data. C error: Expected 81 fields in line 3, saw 83
```

In [22]:
```python
#Inspect the first five lines of the 2018_SCADA_Demands.csv
# This displays the first five lines, including commas and quotation marks, so we c

from pathlib import Path

file_path = project_dir / "2018_SCADA_Demands.csv"

with open(file_path, "r", encoding="utf-8-sig", errors="replace") as f:
    for line_number in range(1, 6):
        line = f.readline()
        if not line:
            break
        print(f"\n--- Line {line_number} ---")
        print(repr(line[:1000]))
```

```
--- Line 1 ---
'Timestamp;n1;n2;n3;n4;n6;n7;n8;n9;n10;n11;n13;n16;n17;n18;n19;n20;n21;n22;n23;n24;n
25;n26;n27;n28;n29;n30;n31;n32;n33;n34;n35;n36;n39;n40;n41;n42;n43;n44;n45;n343;n34
4;n345;n346;n347;n349;n350;n351;n352;n353;n354;n355;n356;n357;n358;n360;n361;n362;n3
64;n365;n366;n367;n368;n369;n370;n371;n372;n373;n374;n375;n376;n377;n378;n379;n381;n
382;n383;n384;n385;n386;n387;n388;n389\n'

--- Line 2 ---
'2018-01-01 00:00:00;40,83;112,23;96,71;131,29;166,18;81,92;249,95;489,63;58,83;495,
98;266,97;99,39;251,8;70,03;85,34;48,12;53,85;135,47;194,87;39,39;989,73;32,63;66,1
5;540,76;65,42;134,28;99,56;67,56;103,78;0;41,65;210,53;10,8;323,74;74,67;111,3;19,7
3;24,96;98,54;94,25;120,71;41,93;200,02;0,64;27,24;83,99;329,38;93,91;359,98;126,69;
0;118,26;90,4;241,93;250,67;137,42;6,22;18,58;114,43;56,9;77,96;75,7;96,52;206,12;8
5,18;47,01;10,37;16,38;60,49;58,27;318,46;4,35;33,41;150,82;67,61;65,82;65,25;57,39;
214,2;81,49;256,5;92,14\n'

--- Line 3 ---
'2018-01-01 00:05:00;40,2;166,05;239,53;136,21;154,48;372,14;190,56;171,6;38,02;142,
36;204,21;63,41;234,08;61,37;81,81;113,65;15,46;114,08;66,72;20,63;1045,72;27,48;31,
47;300,57;32,92;332,56;69,52;35,38;2,95;40,9;30,05;147,01;10,35;317,53;106,94;159,3
9;54,97;36,93;232,74;149,76;172,87;32,67;173,1;0,74;39,01;73,9;167,83;8,06;181,16;11
1,02;158,98;42,96;59,69;115,08;245,86;48,16;2,59;84,39;63,07;60,23;43,29;112,01;123,
73;382,75;54,35;35,96;24,49;80,9;33,34;86,58;191,64;4,61;78,9;97,46;95,29;50,18;81,2
9;49,67;109,14;104,46;394,48;203,61\n'

--- Line 4 ---
'2018-01-01 00:10:00;38,06;121,91;69,12;131;160,1;320,79;242,07;39,35;52,79;225,65;9
5,82;64,72;242,59;19,32;72,39;144,42;24,5;83,14;91,22;2,53;1131,46;5,36;42,73;237,6
9;21,48;95,97;81,71;4,35;89,3;25,19;10,89;172,8;9,16;159,18;29,21;43,53;84,38;27,11;
295,74;112,56;47,21;27,26;33,79;0,82;10,66;149,12;439,5;31,6;118,2;34,96;97,92;104,
2;90,4;156,28;123,25;11,04;5,14;72,74;130,99;118,6;104,62;82,23;111,72;280,28;55,47;
16,87;31,12;135,79;69,25;103,6;286,08;9,07;100,26;135,33;178,1;63,74;252,94;9,69;28
5,82;94,32;337,33;93,11\n'

--- Line 5 ---
'2018-01-01 00:15:00;37,24;33,15;139,01;155,02;244,34;335,55;128,87;252,95;34,28;0;1
16,47;54,73;370,23;19,52;110,91;116,38;0;152,27;58,62;0,3;1061,15;9,31;78,02;256,35;
43,43;193;33,78;0,51;171,78;68,61;62,6;71,43;14,03;263,11;119,24;177,73;103,03;7,37;
238,32;37,75;192,75;23,5;125,98;0,85;43,5;5,92;282,42;22,43;239;35,31;266,7;82,73;0;
285,33;203,73;70,99;3,03;76,09;40,74;96,36;103,41;22,36;67,84;203,23;46,91;20,51;25,
08;45,86;21,54;112,84;48,71;7,37;80,8;87,88;51,6;33,93;139,25;36,15;183,66;57,27;26
8,74;96,19\n'
```

In [23]:
```python
# Check the field counts

import csv

with open(file_path, "r", encoding="utf-8-sig", newline="") as f:
    reader = csv.reader(f)
    for row_number, row in zip(range(1, 6), reader):
        print(f"Line {row_number}: {len(row)} fields")
```

```
Line 1: 1 fields
Line 2: 81 fields
Line 3: 83 fields
Line 4: 82 fields
Line 5: 78 fields
```

In [ ]:
```
# We've found the cause. Your 2018_SCADA_Demands.csv uses semicolons to separate co
# For example: Timestamp;n1;n2;n3;... – semicolons separate the columns.
# 40,83 – the comma represents the decimal in the value 40.83.
# Your original code used pd.read_csv(path), which assumes comma-separated columns
# That makes Pandas split decimal values such as 40,83 into separate fields, causin
# The file's inconsistent field counts are therefore explained by the wrong delimit
# We should read it using sep=";" and decimal="," instead.
```

In [24]:
```
# LOAD THE DEMANDS_2018 FILE CORRECTLY
import pandas as pd

file_path = r'C:\Users\HP\Downloads\SCADA Water Analytics\2018_SCADA_Demands.csv'
demands_2018 = pd.read_csv(
    file_path,
    sep=';',
    decimal=','
)
```

In [25]:
```
demands_2018.head(3)
```

Out[25]:

	Timestamp	n1	n2	n3	n4	n6	n7	n8	n9	n10	...	n3
0	2018-01-01 00:00:00	40.83	112.23	96.71	131.29	166.18	81.92	249.95	489.63	58.83	...	33.
1	2018-01-01 00:05:00	40.20	166.05	239.53	136.21	154.48	372.14	190.56	171.60	38.02	...	78.
2	2018-01-01 00:10:00	38.06	121.91	69.12	131.00	160.10	320.79	242.07	39.35	52.79	...	100.

3 rows × 83 columns

In [26]:
```
demands_2018.shape

# 105120 rows and 83 columns
```

Out[26]:
```
(105120, 83)
```

In [27]:
```
demands_2018.dtypes
```

Out[27]:
```
Timestamp        str
n1           float64
n2           float64
n3           float64
n4           float64
              ...
n385         float64
n386         float64
n387         float64
n388         float64
n389         float64
Length: 83, dtype: object
```

In [57]:
```
# Convert the Timestamp dtype to datetime
demands_2018["Timestamp"] = pd.to_datetime(
    demands_2018["Timestamp"],
    errors="coerce"
)
```

In [58]:
```
demands_2018.dtypes
```

Out[58]:
```
Timestamp    datetime64[us]
n1                  float64
n2                  float64
n3                  float64
n4                  float64
                    ...
n385                float64
n386                float64
n387                float64
n388                float64
n389                float64
Length: 83, dtype: object
```

In [50]:
```
demands_2018.isnull().sum()
```

Out[50]:
```
Timestamp    0
n1           0
n2           0
n3           0
n4           0
            ..
n385         0
n386         0
n387         0
n388         0
n389         0
Length: 83, dtype: int64
```

In [ ]:
```
# 2018 SCADA Demands file has no missing values
```

In [56]:
```
demands_2018.duplicated().sum()
```

Out[56]:
```
np.int64(0)
```

In [ ]:
```
# 2018 SCADA Demands file has no duplicated values
```

In [28]:
```
# LOAD LEVELS_2018

import pandas as pd

file_path = r'C:\Users\HP\Downloads\SCADA Water Analytics\2018_SCADA_Levels.csv'

levels_2018 = pd.read_csv(
    file_path,
    sep=';',
    decimal=','
)
```

In [29]:
```
levels_2018.head()
```

Out[29]:

	Timestamp	T1
0	2018-01-01 00:00:00	3.50
1	2018-01-01 00:05:00	3.51
2	2018-01-01 00:10:00	3.53
3	2018-01-01 00:15:00	3.54
4	2018-01-01 00:20:00	3.56

In [30]:
```
levels_2018.shape

# 105120 rows and 2 columns
```

Out[30]:
```
(105120, 2)
```

In [31]:
```
levels_2018.dtypes
```

Out[31]:
```
Timestamp        str
T1           float64
dtype: object
```

In [59]:
```
# Convert the Timestamp dtype to datetime
levels_2018["Timestamp"] = pd.to_datetime(
    levels_2018["Timestamp"],
    errors="coerce"
)
```

In [60]:
```
levels_2018.dtypes
```

Out[60]:
```
Timestamp    datetime64[us]
T1                  float64
dtype: object
```

In [49]:
```
levels_2018.isnull().sum()
```

Out[49]:
```
Timestamp    0
T1           0
dtype: int64
```

In [ ]:
```
# 2018 SCADA Levels file has no missing values
```

In [55]:
```
levels_2018.duplicated().sum()
```

Out[55]:
```
np.int64(0)
```

In [ ]:
```
# 2018 SCADA Levels file has no duplicated values
```

In [32]:
```
# LOAD PRESSURES_2018

import pandas as pd

file_path = r'C:\Users\HP\Downloads\SCADA Water Analytics\2018_SCADA_Pressures.csv'

pressures_2018 = pd.read_csv(
    file_path,
    sep=';',
    decimal=','
)
```

In [33]:
```
pressures_2018.head()
```

Out[33]:

	Timestamp	n1	n4	n31	n54	n105	n114	n163	n188	n215	...	n549	n613
0	2018-01-01 00:00:00	28.92	33.87	37.13	37.13	50.52	53.99	52.56	55.32	39.09	...	54.80	56.16
1	2018-01-01 00:05:00	28.94	33.89	37.16	37.23	50.60	54.09	52.67	55.42	39.10	...	54.90	56.26
2	2018-01-01 00:10:00	28.96	33.91	37.18	37.21	50.57	54.05	52.64	55.39	39.09	...	54.87	56.24
3	2018-01-01 00:15:00	28.98	33.93	37.19	37.27	50.60	54.08	52.70	55.45	39.09	...	54.92	56.30
4	2018-01-01 00:20:00	28.98	33.93	37.19	37.14	50.53	54.00	52.57	55.32	39.09	...	54.80	56.17

5 rows × 34 columns

In [35]:
```
pressures_2018.shape

# 105120 rows and 34 columns
```

Out[35]:
```
(105120, 34)
```

In [36]:
```
pressures_2018.dtypes
```

Out[36]:
```
Timestamp        str
n1           float64
n4           float64
n31          float64
n54          float64
n105         float64
n114         float64
n163         float64
n188         float64
n215         float64
n229         float64
n288         float64
n296         float64
n332         float64
n342         float64
n410         float64
n415         float64
n429         float64
n458         float64
n469         float64
n495         float64
n506         float64
n516         float64
n519         float64
n549         float64
n613         float64
n636         float64
n644         float64
n679         float64
n722         float64
n726         float64
n740         float64
n752         float64
n769         float64
dtype: object
```

In [61]:
```
# Convert the Timestamp dtype to datetime
pressures_2018["Timestamp"] = pd.to_datetime(
    pressures_2018["Timestamp"],
    errors="coerce"
)
```

In [62]:
```
pressures_2018.dtypes
```

Out[62]:
```
Timestamp    datetime64[us]
n1                  float64
n4                  float64
n31                 float64
n54                 float64
n105                float64
n114                float64
n163                float64
n188                float64
n215                float64
n229                float64
n288                float64
n296                float64
n332                float64
n342                float64
n410                float64
n415                float64
n429                float64
n458                float64
n469                float64
n495                float64
n506                float64
n516                float64
n519                float64
n549                float64
n613                float64
n636                float64
n644                float64
n679                float64
n722                float64
n726                float64
n740                float64
n752                float64
n769                float64
dtype: object
```

In [48]:
```
pressures_2018.isnull().sum()
```

Out[48]:
```
Timestamp    0
n1           0
n4           0
n31          0
n54          0
n105         0
n114         0
n163         0
n188         0
n215         0
n229         0
n288         0
n296         0
n332         0
n342         0
n410         0
n415         0
n429         0
n458         0
n469         0
n495         0
n506         0
n516         0
n519         0
n549         0
n613         0
n636         0
n644         0
n679         0
n722         0
n726         0
n740         0
n752         0
n769         0
dtype: int64
```

In [ ]:
```
# 2018 SCADA Pressures file has no missing values
```

In [54]:
```
pressures_2018.duplicated().sum()
```

Out[54]:
```
np.int64(0)
```

In [ ]:
```
# 2018 SCADA Pressures file has no duplicated values
```

In [37]:
```
# LOAD FLOWS_2018

import pandas as pd

file_path = r'C:\Users\HP\Downloads\SCADA Water Analytics\2018_SCADA_Flows.csv'

flows_2018 = pd.read_csv(
    file_path,
    sep=';',
    decimal=','
)
```

In [38]:
```
flows_2018.head()
```

Out[38]:

	Timestamp	p227	p235	PUMP_1
0	2018-01-01 00:00:00	77.77	83.93	44.04
1	2018-01-01 00:05:00	72.51	76.34	44.06
2	2018-01-01 00:10:00	71.54	78.68	44.06
3	2018-01-01 00:15:00	69.85	75.50	44.07
4	2018-01-01 00:20:00	77.12	82.99	44.03

In [39]:
```
flows_2018.shape

# 105120 rows and 4 columns
```

Out[39]:
```
(105120, 4)
```

In [40]:
```
flows_2018.dtypes
```

Out[40]:
```
Timestamp        str
p227         float64
p235         float64
PUMP_1       float64
dtype: object
```

In [63]:
```
# Convert the Timestamp dtype to datetime
flows_2018["Timestamp"] = pd.to_datetime(
    flows_2018["Timestamp"],
    errors="coerce"
)
```

In [64]:
```
flows_2018.dtypes
```

Out[64]:
```
Timestamp    datetime64[us]
p227                float64
p235                float64
PUMP_1              float64
dtype: object
```

In [47]:
```
flows_2018.isnull().sum()
```

Out[47]:
```
Timestamp    0
p227         0
p235         0
PUMP_1       0
dtype: int64
```

In [ ]:
```
# 2018_SCADA_FLOWS FILE HAS NO MISSING VALUES
```

In [53]:
```
flows_2018.duplicated().sum()
```

Out[53]:
```
np.int64(0)
```

In [ ]:
```
# 2018_SCADA_FLOWS FILE HAS NO DUPLICATED VALUES
```

In [41]:
```
# load the 2018 LEAKAGES FILE

import pandas as pd

file_path = r'C:\Users\HP\Downloads\SCADA Water Analytics\2018_Leakages.csv'

leakages_2018 = pd.read_csv(
    file_path,
    sep=';',
    decimal=','
)
```

In [42]:
```
leakages_2018.head()
```

Out[42]:

	Timestamp	p31	p158	p183	p232	p257	p369	p427	p461	p538	p628	p654	p67
0	2018-01-01 00:00:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
1	2018-01-01 00:05:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
2	2018-01-01 00:10:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
3	2018-01-01 00:15:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
4	2018-01-01 00:20:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0

In [43]:
```
leakages_2018.shape

# 105120 rows and 15 columns
```

Out[43]:
```
(105120, 15)
```

In [44]:
```
leakages_2018.dtypes
```

Out[44]:
```
Timestamp        str
p31          float64
p158         float64
p183         float64
p232         float64
p257         float64
p369         float64
p427         float64
p461         float64
p538         float64
p628         float64
p654         float64
p673         float64
p810         float64
p866         float64
dtype: object
```

In [65]:
```
# Convert the Timestamp dtype to datetime
leakages_2018["Timestamp"] = pd.to_datetime(
    leakages_2018["Timestamp"],
    errors="coerce"
)
```

In [66]:
```
leakages_2018.dtypes
```

Out[66]:
```
Timestamp    datetime64[us]
p31                 float64
p158                float64
p183                float64
p232                float64
p257                float64
p369                float64
p427                float64
p461                float64
p538                float64
p628                float64
p654                float64
p673                float64
p810                float64
p866                float64
dtype: object
```

In [46]:
```
leakages_2018.isnull().sum()
```

Out[46]:
```
Timestamp    0
p31          0
p158         0
p183         0
p232         0
p257         0
p369         0
p427         0
p461         0
p538         0
p628         0
p654         0
p673         0
p810         0
p866         0
dtype: int64
```

In [ ]:
```
# 2018_Leakages file has no missing values
```

In [52]:
```
leakages_2018.duplicated().sum()
```

Out[52]:
```
np.int64(0)
```

In [ ]:
```
# 2018_Leakages file has no duplicated values
```

In [67]:
```
# Export the cleaned leakages_2018 file as a standard CSV file
output_path = 'cleaned_leakages_2018.csv'
leakages_2018.to_csv(output_path, index=False, encoding='utf-8')
print()
```

In [68]:
```
# Export the cleaned Pressures_2018 file as a standard CSV file
output_path = 'cleaned_pressures_2018.csv'
pressures_2018.to_csv(output_path, index=False, encoding='utf-8')
print()
```

In [69]:
```
# Export the cleaned levels_2018 file as a standard CSV file
output_path = 'cleaned_levels_2018.csv'
levels_2018.to_csv(output_path, index=False, encoding='utf-8')
print()
```

In [70]:
```
# Export the cleaned demands_2018 file as a standard CSV file
output_path = 'cleaned_demands_2018.csv'
demands_2018.to_csv(output_path, index=False, encoding='utf-8')
print()
```

In [71]:
```
# Export the cleaned flows_2018 file as a standard CSV file
output_path = 'cleaned_flows_2018.csv'
flows_2018.to_csv(output_path, index=False, encoding='utf-8')
print()
```

In [ ]:
```

```